



Commercial Insurance Policy Comparison & Risk Insights Platform

Business Analysis Case Study | Commercial Insurance | SQL | Tableau | PostgreSQL

Prepared By

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Business Analyst Portfolio



Domain

**Commercial
Insurance**



Role

Business Analyst



Tools

**PostgreSQL
L · SQL · Tableau**

Executive Summary

Business challenge, proposed solution, and key capabilities

Business Challenge

Commercial insurance brokers compare multiple insurer policies manually, leading to lengthy review cycles, inconsistent advisory decisions, and limited visibility into coverage gaps.

Proposed Solution

A centralized platform to standardize policy comparison, identify coverage gaps, support broker recommendations, and maintain an auditable decision trail.

Key Capabilities

Capability	Business Value
Policy Comparison	Faster evaluation of multiple insurer policies
Coverage Gap Detection	Improved risk assessment
Recommendation Engine	Consistent broker advisory decisions
Audit Trail	Governance and compliance
SQL Analytics & Dashboard	Operational visibility and KPI tracking

Business Problem Statement

Current challenges and their business impact

Challenge	Business Impact
Manual policy comparison	Longer turnaround time
Different insurer terminology	Inconsistent policy interpretation
Hidden exclusions	Increased advisory risk
Spreadsheet-based analysis	Limited traceability
No centralized audit trail	Higher E&O exposure

Business Impact

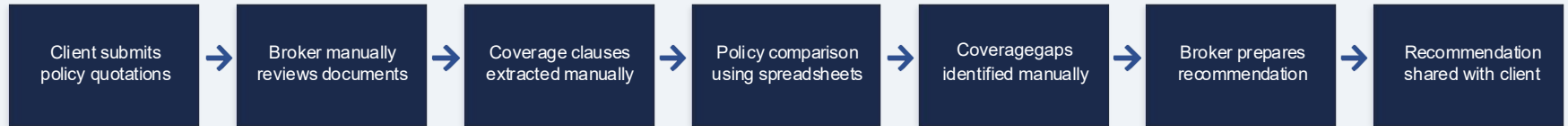
- Time-consuming policy evaluation
- Recommendation inconsistencies
- Difficult coverage gap identification
- Limited operational visibility
- Higher governance & compliance risk

Opportunity

Develop a centralized platform that standardizes policy comparison, improves advisory consistency, and provides complete audit traceability while supporting data-driven decision-making.

Current Process (AS-IS)

Manual policy comparison workflow



Key Process Challenges

Current State	Business Impact
Manual document review	Increased processing time
Spreadsheet-based comparison	Version control issues
Inconsistent clause interpretation	Advisory inconsistencies
Manual coverage gap analysis	Higher risk of oversight
Limited audit trail	Reduced governance visibility

Business Consequences

- 4- 6 hours average policy comparison
- Inconsistent broker recommendations
- Higher Errors & Omissions
- Limited audit readiness
- Reduced operational scalability

Future Process (TO-BE)

Proposed digital workflow with standardized processes



Business Improvements

Current Process	Proposed Process
Manual policy review	Standardized digital workflow
Spreadsheet comparison	Centralized comparison platform
Manual clause interpretation	Standardized clause taxonomy
Manual gap identification	Automated gap highlighting
Limited auditability	Complete audit trail

Expected Business Value

- Faster policy comparison
- Standardized clause evaluation
- Better coverage gap visibility
- Improved operational reporting
- Stronger governance & compliance

Requirement Elicitation

Approach, stakeholders, and key findings

Technique	Stakeholders	Key Output
Workshops	Brokers, Underwriters	Business Requirements
Interviews	Compliance, Operations	Pain Points & Process Gaps
Process Observation	Insurance Brokers	AS-IS Workflow
Document Analysis	Policy Documents	Business Rules & Clause Mapping
Data Analysis	Operations Team	KPIs & Performance Baseline

Key Findings

- Manual policy comparison increased turnaround time
- Different insurer terminology caused inconsistent policy interpretation
- Limited audit traceability increased governance risk

Stakeholder Analysis

Stakeholder matrix and engagement priorities

Stakeholder	Responsibility	Priority
Insurance Broker	Policy comparison & recommendations	High
Enterprise Client	Review recommendations	High
Underwriting Team	Validate policy terms	High
Compliance Team	Governance & audit	High
Product Manager	Prioritize roadmap	Medium
Engineering Team	Solution implementation	Medium
QA Team	Business validation	Medium

Stakeholder Engagement Matrix

High Influence	
Brokers Compliance	Product Manager Enterprise Client
Engineering	QA Team
Low Influence	

Functional Requirements

Core functional modules and key requirements



Policy Upload

Upload and validate policy documents



Clause Extraction

Extract coverage clauses and exclusions



Policy Comparison

Side-by-side comparison of multiple policies



Recommendation Workflow

Generate and review broker recommendations



Reporting & Audit

Operational dashboards and audit trail

Key Functional Requirements

- Multi-policy upload
- Clause standardization
- Coverage gap identification
- Recommendation generation
- Audit logging

Non-Functional Requirements

System quality attributes and compliance needs

Category	Requirement
Performance	Policy comparison within 60 seconds
Availability	99.5% availability
Security	Role-Based Access Control (RBAC)
Auditability	Complete recommendation traceability
Scalability	Support concurrent policy comparisons
Reliability	Log failed uploads and processing errors
Compliance	Protect sensitive client information

Business Importance

- Reliable system performance
- Secure access to policy data
- Regulatory compliance
- Operational stability

Business Rules

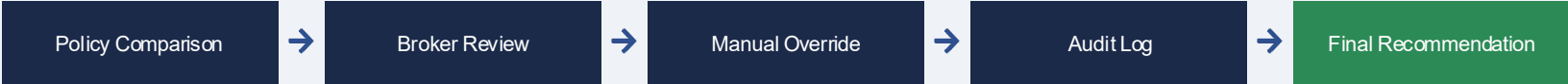
Key rules and governance principles

Rule	Description
BR-01	Compare only policies of the same product category
BR-02	Only brokers can approve final recommendations
BR-03	Manual overrides require mandatory justification
BR-04	All key actions are recorded in the audit trail
BR-05	Critical exclusions must be highlighted
BR-06	Revised policy documents require a new version

Governance Principles

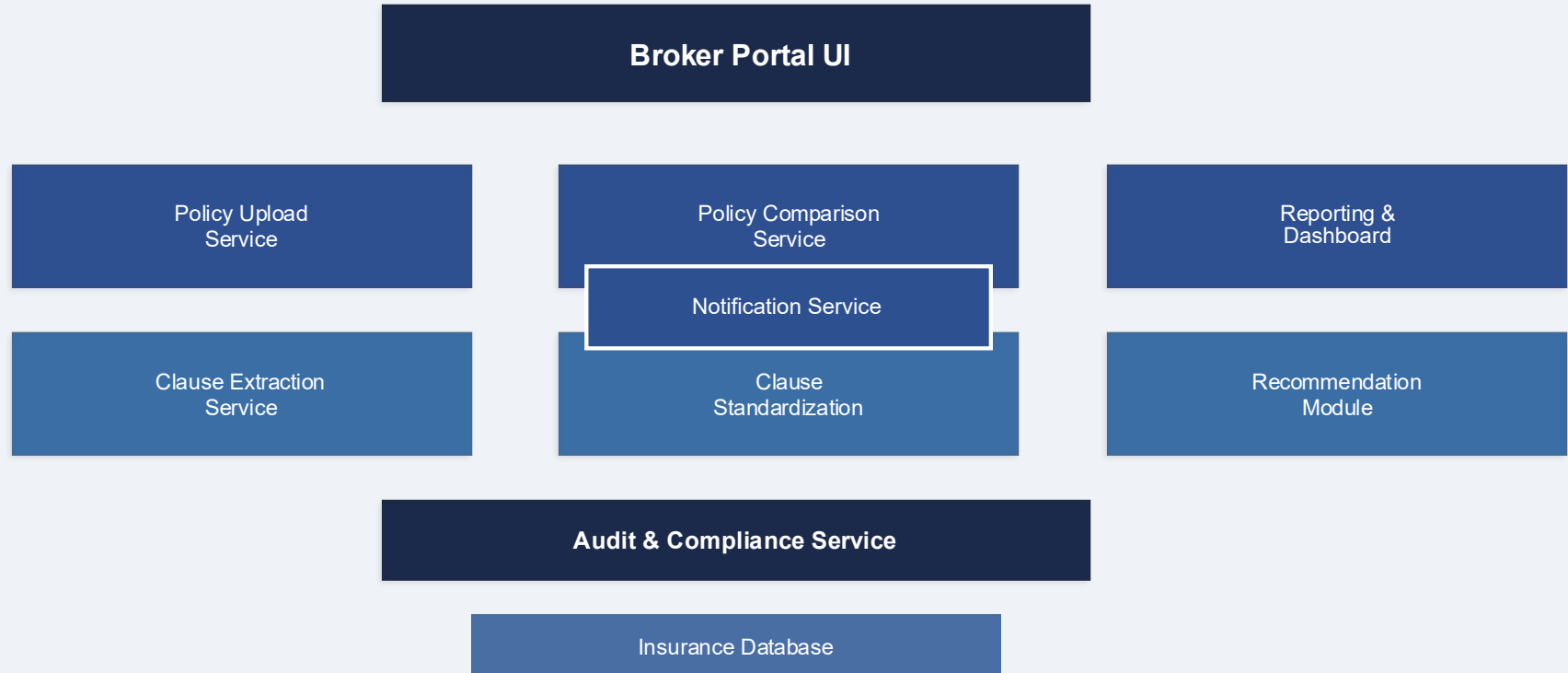
- Standardized policy comparison
- Controlled recommendation workflow
- Full audit traceability
- Role-based authorization

Governance Flow



Solution Architecture

High-level platform architecture and core components



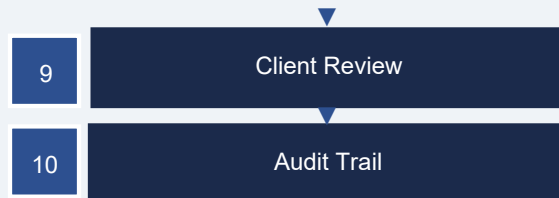
Business Process Flow

End-to-end process from upload to audit



Business Outcomes

- Reduced manual effort
- Standardized comparison
- Faster advisory decisions
- Better compliance

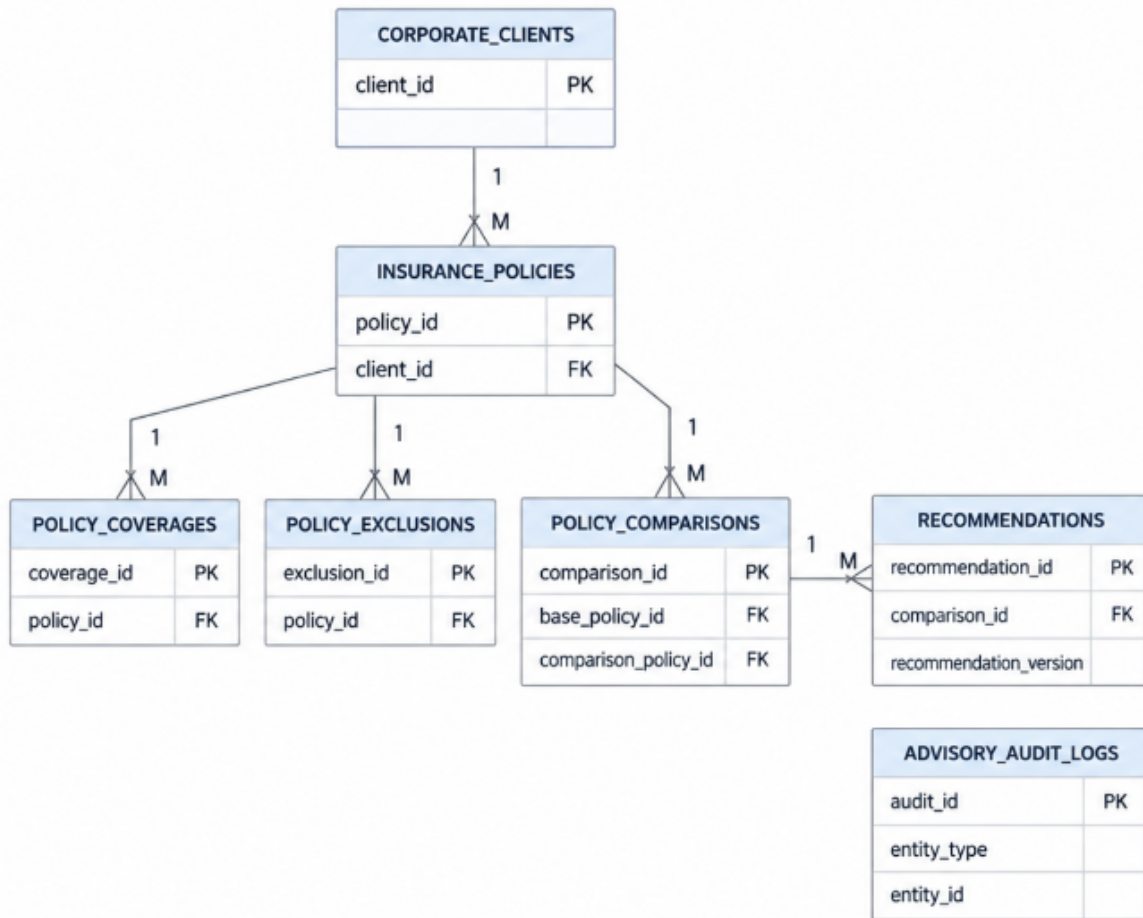


Data Model

Entity Relationship
Diagram and key entities

Key Entities

Entity	Purpose
Corporate Clients	Client information
Insurance Policies	Policy records
Policy Coverages	Coverage details
Policy Exclusions	Risk exclusions
Recommendations	Broker decisions
Audit Logs	Governance history



User Stories & Integration

Epics, user stories, and API integrations

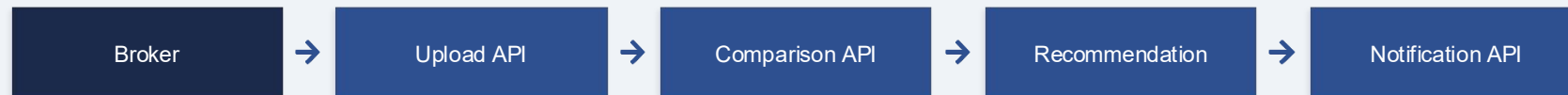
Key User Stories

Epic	Representative Story
Policy Upload	Upload policy documents
Policy Comparison	Compare insurer policies
Recommendation	Generate recommendation
Governance	Maintain audit trail

API Integrations

API	Purpose
Upload API	Upload policy documents
Comparison API	Compare policies
Notification API	Notify broker on completion

Integration Flow



SQL Analytics

Key business queries and analytical capabilities

Key Business Queries

Analytics	Business Purpose
Policy Comparison Turnaround	Monitor comparison processing time
Coverage Exclusion Density	Identify high-risk exclusions
Broker Productivity	Track comparison workload
Recommendation Turnaround	Monitor recommendation completion time
Audit Activity	Monitor governance actions

SQL Capabilities

- CTEs
- JOINS
- Aggregations
- Conditional Aggregation
- Timestamp Calculations
- Business KPI Reporting

Example: Policy Comparison Turnaround Query

```
SELECT COUNT(comparison_id) AS total_comparisons_completed, ROUND(AVG( EXTRACT(EPOCH FROM  
(comparison_completed_at - comparison_started_at)) / 3600.0 ), 2) AS average_comparison_turnaround_hours  
FROM PolicyComparisons WHERE comparison_status = 'Completed';
```

Tableau Dashboard

Executive dashboard highlights and business insights



Comparison Turnaround

Monitor policy comparison processing time



Exclusion & Risk Analysis

Identify exclusion concentration and critical policy risks



Audit & Broker Activity

Track advisory actions and comparison activity



Recommendation Turnaround

Monitor time required to complete broker recommendations

Business Insights

- Operational performance tracking
- Coverage risk visibility
- Audit activity monitoring
- Executive decision support

Business KPIs

Success metrics and expected outcomes

Faster

Policy Comparison
Turnaround

Operational efficiency

Better

Coverage Gap
Detection

Improved
recommendations

Higher

Broker
Productivity

Improved throughput

100%

Audit
Audit Traceability

Governance visibility

Timely

Recommendation
Turnaround

Monitor completion time

Expected Outcomes

- Faster policy comparison
- Reduced manual effort
- Improved governance
- Standardized advisory process

Business Impact

Before vs. after comparison and expected business value

Before	After
Manual comparison	Standardized workflow
Spreadsheet tracking	Centralized platform
Manual gap analysis	Structured comparison
Limited audit trail	Complete traceability

Expected Business Value



Policy Comparison Time

4–6 hrs → Target: <1 hr



Manual Effort

Target reduction: 60%



Audit Traceability

Key actions logged



Recommendation Consistency

Standardized & Traceable

Project Deliverables

Documentation, technical assets, and skills demonstrated

Documentation

- Business Analysis Document
- Functional & Non-Functional Requirements
- User Stories
- API Contracts
- UAT

Technical Assets

- ERD
- PostgreSQL Schema
- Sample Dataset
- SQL Analytics
- Tableau Dashboard
- GitHub Repository

Skills Demonstrated

- Business Analysis
- SQL
- Tableau
- Data Modelling
- Process Design

Project Summary

Manual commercial insurance policy comparison resulted in inconsistent recommendations, longer turnaround times, and limited audit traceability.

A centralized Policy Comparison & Risk Insights Platform supporting standardized comparisons, governance, SQL analytics, and executive dashboards.

Key Outcomes



Standardized policy
comparison



Improved operational
efficiency



Enhanced governance



Better business visibility

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Thank You